

Lesson 11: Telescoping Series

Assessment Criteria Rubric

- **Identification (15%):** Correct identification of the series as telescoping, often by expanding terms or rewriting.
- **Decomposition (25%):** Accurate use of partial fraction decomposition (e.g., $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$) to form a difference of two terms.
- **Partial Sum Calculation (S_n) (30%):** Correct derivation of the n -th partial sum S_n , accurately listing initial terms and final terms, and showing how intermediate terms cancel.
- **Convergence and Limit (20%):** Evaluation of $\lim_{n \rightarrow \infty} S_n$. The series converges if the limit is a finite number s , and diverges if the limit is $\pm\infty$ or does not exist.
- **Notation and Rigor (10%):** Proper mathematical notation, including sigma notation ($\sum$) and limit notation ($\lim_{n \rightarrow \infty}$).